# “DPPL”




**OLEH:**

**Stefanus Andri Hendrawan 103072400115**

**Fatih Khairu Alfifajri 103072400105**

**Irham Maulana .A. 103072400063**

**Alif Lutfan .A. 1030724001634**

**FAKULTAS INFORMATIKA**

**PROGRAM STUDI INFORMATIKA**

**TELKOM UNIVERSITY SURABAYA**

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# DOKUMEN 2: DPPL (Dokumen Perancangan Perangkat Lunak)

Identifikasi Dokumen: DPPL-lMPAL-2026-01

Nama Sistem: SmartKasir (Point of Sale System)

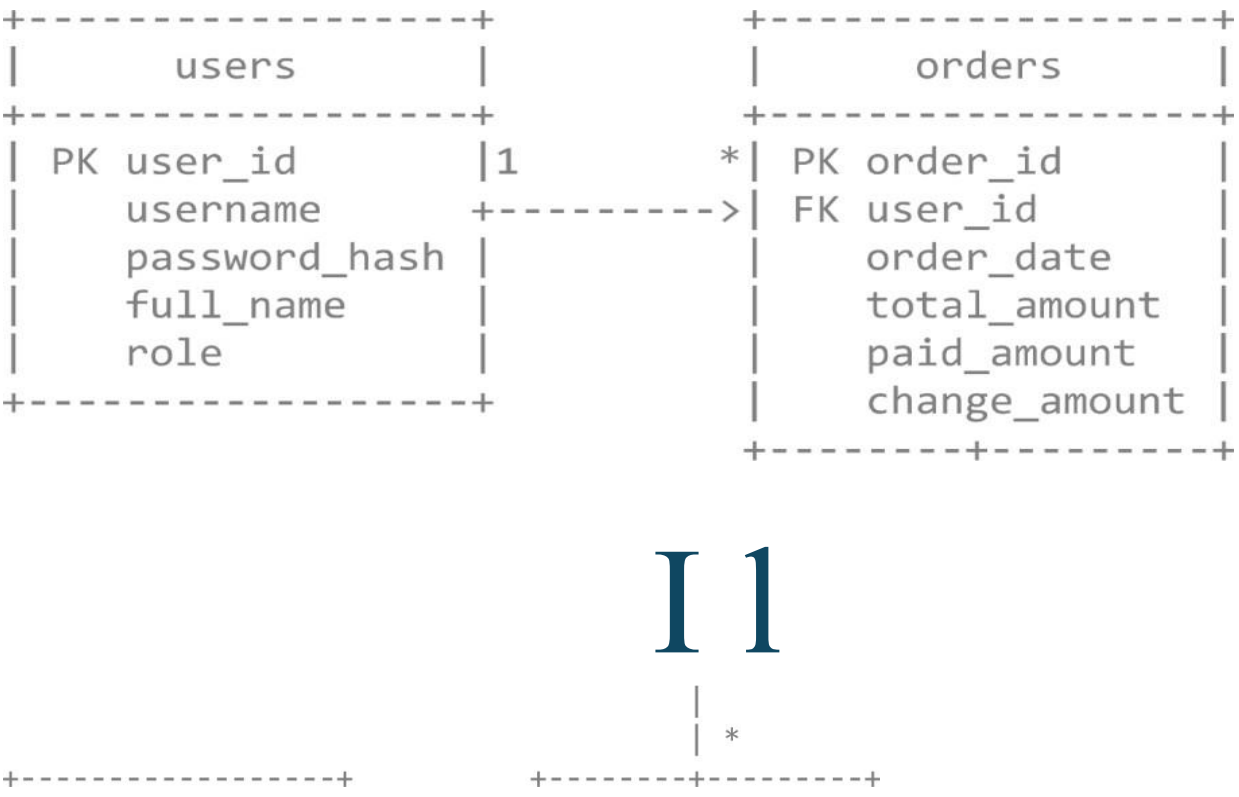
## 1. DESAIN ARSITEKTUR SISTEM

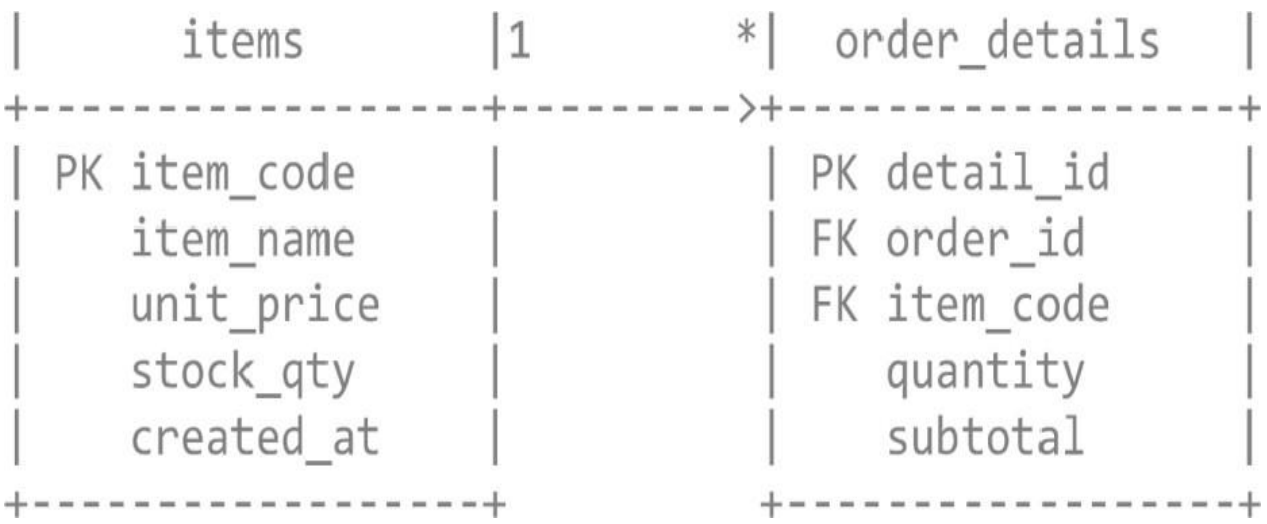
Sistem dirancang menggunakan arsitektur berlapis (Layered Pattern / MVC Pattern):

1. Presentation Layer (View): Menangani interaksi langsung dengan kasir/admin (Form Login, Layar Transaksi Kasir, Form Manajemen Barang).
2. Business Logic Layer (Controller / Service): Memvalidasi masukan kasir, aturan bisnis perhitungan diskon/pajak (jika ada), kalkulasi kembalian, orkestrasi pemotongan stok, dan service rendering struk belanja PDF.
3. Data Access Layer (Model / DAO): Bertanggung jawab atas kueri SQL transaksi (Data Definition & Manipulation) ke database engine (MySQL / PostgreSQL / SQLite).

## 2. DESAIN BASIS DATA (PHYSICAL DATA MODEL)

### Skema Relasi Database






## Kamus Data Tabel

Tabel: users

- user id (INT, Primary Key, Auto Increment)
- username (VARCHAR(50), Unique, Not Null)
- password_hash (VARCHAR(255), Not Null)
- full name (VARCHAR(IOO), Not Null)
- role (ENUM('ADMIN', 'CASHIER'), Not Null)

Tabel: items

- item code (VARCHAR(30), Primary Key) • item name (VARCHAR(150), Not Null)
- unit_ price (DECIMAL(12,2), Not Null)
- stock_qty (INT, Not Null, Default: 0) • created at (TIMESTAMP, Default: CURRENT_TIMESTAMP)

Tabel: orders

- order id (VARCHAR(35), Primary Key)— Format: TRX-YYYYMMDD-XXXX • order date (DATETIME, Not Null) • user id (INT, Foreign Key users. user_id)
- total amount DECIMAL(12,2), Not Null) • paid_amount(DECIMAL(12,2), Not Null) • change_amount(DECIMAL(12,2),Not Null)
- detail id (INT, Primary Key, Auto Increment) • order id (VARCHAR(35), Foreign Key + orders. order_id ON DELETE CASCADE)
    - item code (VARCHAR(30), Foreign Key + items . item_code) • quantity (INT, Not Null)
- subtotal (DECIMAL(12,2), Not Null)

# 3. DESAIN KELAS DAN MODUL PROGRAM (CLASS DIAGRAM SPECIFICATION)

## 3.1 Entitas Model

- User: Atribut (userId, username, passwordHash, role).
- Item: Atribut (itemCode, itemName, unitPrice, stockQty).
- Order: Atribut (orderId, orderDate, userId, itemsList, totalAmount, paidAmount, changeAmount).
- OrderDetai1: Atribut (detailId, orderId, itemCode, quantity, subtotal).

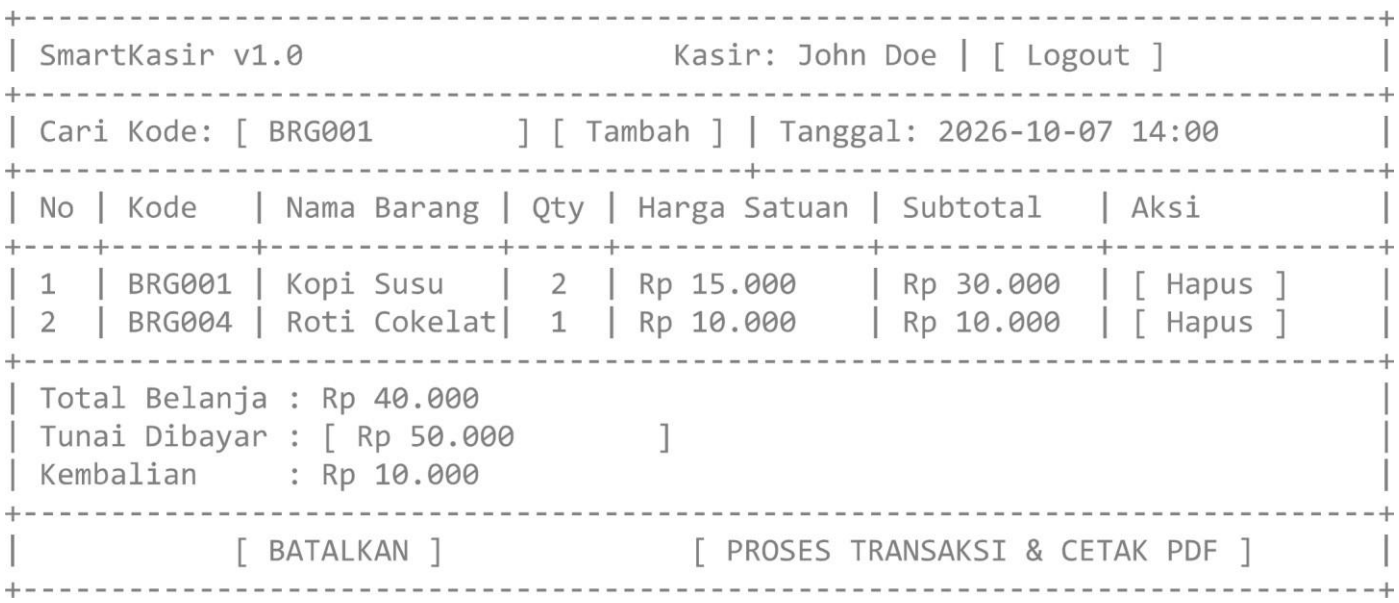
## 3.2 Service & Controller

- AuthService:
- login(username, rawPassword) : UserSession o logout(): void o hashPassword(password) : String
- ProductService:
- get Item(itemCode) : Item o updateStock(itemCode, deltaQty) : boolean o saveProduct(Item) : boolean
- TransactionService:
- calculateSubtota1(unitPrice, qty): double o calculateGrandTota1(cartItems): double o validatePayment(grandTota1, paidAmount) : boolean

- calculateChange(grandTota1, paidAmount) : double
- commit Transaction (Order) : boolean (Menjalankan insert ke orders, order details, dan update items . stock_qty dalam satu transaksi database).
- PdfReceiptService:
- generateReceiptPdf(orderId) : File (Mengisi template struk: daftar item, total, nominal bayar, kembalian, nama kasir, dan waktu transak

## 4. PERANCANGAN ANTARMUKA PENGGUNA (UI WIREFRAME LAYOUT)

### Layout 1: Layar Kasir / POS (Fokus Sprint 2)




## 5. STRUKTUR FOLDER REPOSITORI GIT

```
smartkasir/
├── docs/
│   ├── SKPL_SmartKasir.md
│   ├── DPPL_SmartKasir.md
│   ├── Software_Test_Document.md    # Dikerjakan di Sprint 3
│   └── User_Manual.md               # Dikerjakan di Sprint 3
├── src/                             # Source code aplikasi
│   ├── config/                      # Koneksi database & environment
│   ├── controllers/                 # Logika alur proses & routing
│   ├── models/                      # Definisi entity & query database
│   ├── views/                       # File UI (GUI/HTML/Template)
│   └── utils/                       # Generator PDF, fungsi enkripsi hash
├── database/
│   └── migrations.sql               # Script DDL pembuatan tabel
└── README.md
```

# SEKIAN DAN TERIMA KASIH